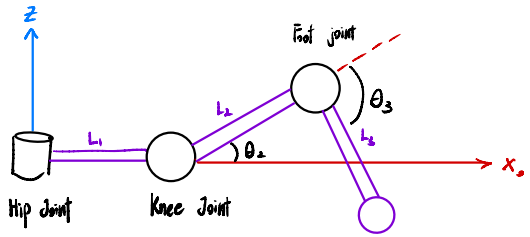
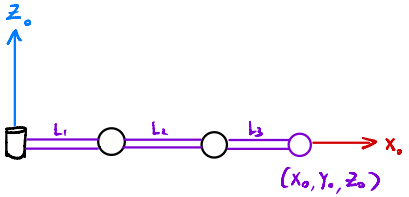
Math calculation :

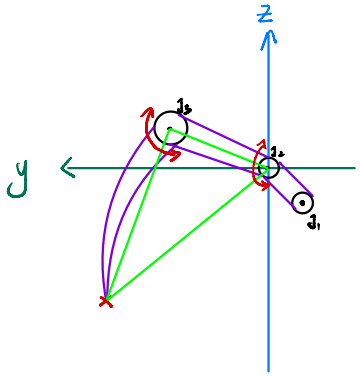
Hexapod Single leg Inverse Kinematic :



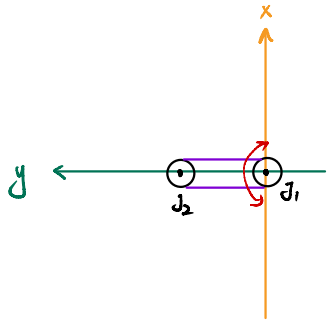
Frame 0 is static frame



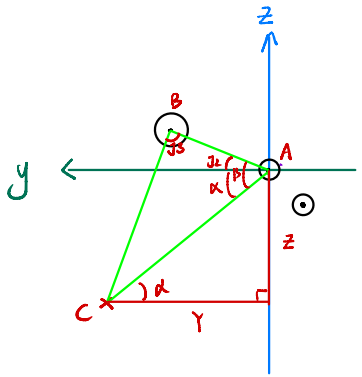
Side View :



Tope view :



Side View :



AB, BC is known.

(Y,Z) : desired foot position

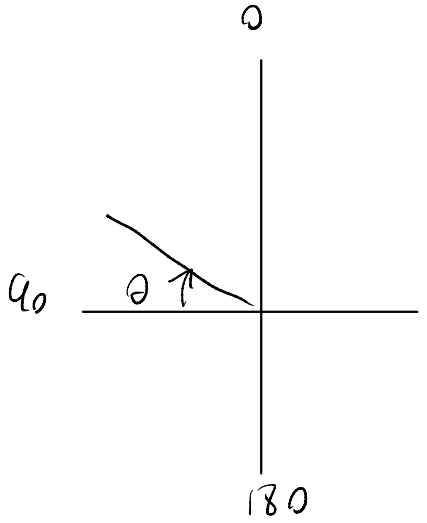
$\Rightarrow AC = \sqrt{y^2 + z^2}$

$J3 = \cos^{-1}\left(\frac{AB^2 + BC^2 - AC^2}{2 \cdot AB \cdot BC}\right)$

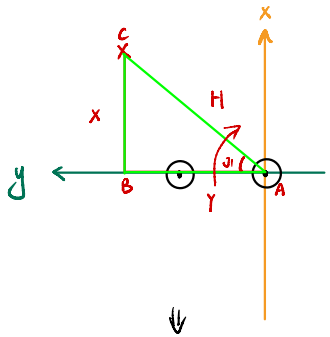
$\beta = \cos^{-1}\left(\frac{AB^2 + AC^2 - BC^2}{2 \cdot AB \cdot AC}\right)$

$\alpha = \tan^{-1}\left(\frac{z}{Y}\right)$

$J2 = \beta - \alpha$



Top view :

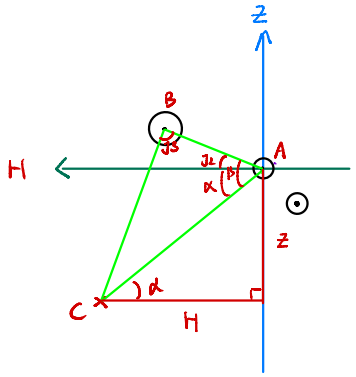


$J1 = \tan^{-1}\left(\frac{X}{Y}\right)$

$AB = \sqrt{X^2 + Y^2}$

Why we need to change J2, J3 calculation?

Ans: beco2 when we doing rotation. J2,J3 not longer using Y as the length. Instead, we use H.



AB, BC is known.

(Y,Z) : desired foot position

$\Rightarrow AC = \sqrt{H^2 + z^2}$

$J3 = \cos^{-1}\left(\frac{AB^2 + BC^2 - AC^2}{2 \cdot AB \cdot BC}\right)$

$\beta = \cos^{-1}\left(\frac{AB^2 + AC^2 - BC^2}{2 \cdot AB \cdot AC}\right)$

$\alpha = \tan^{-1}\left(\frac{z}{H}\right)$

$J2 = \beta - \alpha$